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Session: 2023-2027

Semester: 5th

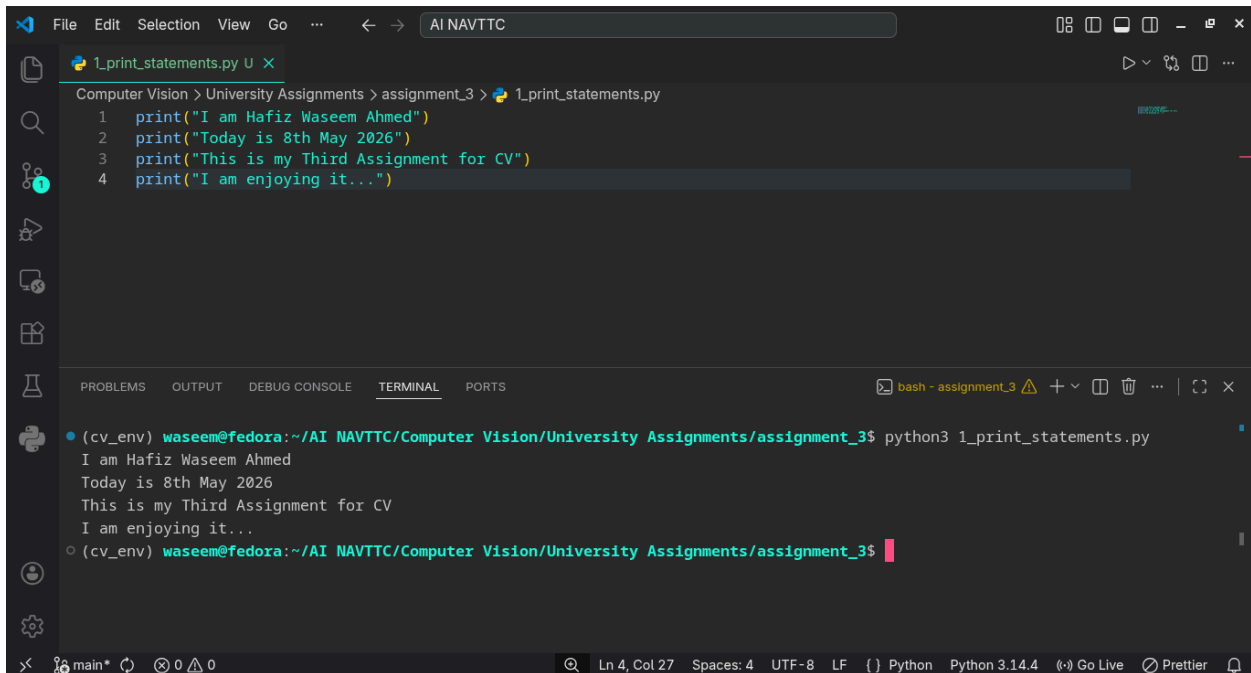
Subject: Computer Vision

Submitted to: Dr. Abdul Basit

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Computer Vision Assignment 3

Task 1: Print Multiple Statements using `print()` function



```
File Edit Selection View Go ... AI NAVTTC
```

```
1_print_statements.py U X
```

```
Computer Vision > University Assignments > assignment_3 > 1_print_statements.py
```

```
1 print("I am Hafiz Waseem Ahmed")
2 print("Today is 8th May 2026")
3 print("This is my Third Assignment for CV")
4 print("I am enjoying it...")
```

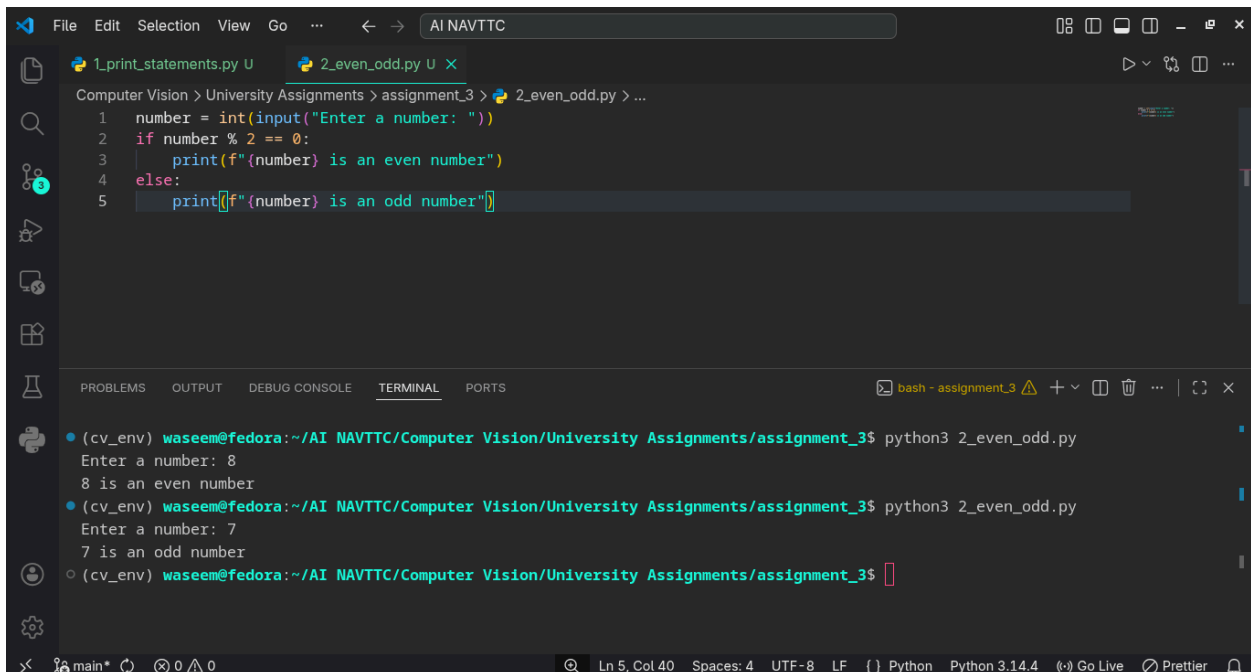
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
```

```
bash - assignment_3
```

```
(cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$ python3 1_print_statements.py
I am Hafiz Waseem Ahmed
Today is 8th May 2026
This is my Third Assignment for CV
I am enjoying it...
(cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$
```

Ln 4, Col 27 Spaces: 4 UTF-8 LF {} Python Python 3.14.4 Go Live Prettier

Task 2: Print if a number is Even or Odd:



```
File Edit Selection View Go ... AI NAVTTC
```

```
1_print_statements.py U 2_even_odd.py U X
```

```
Computer Vision > University Assignments > assignment_3 > 2_even_odd.py > ...
```

```
1 number = int(input("Enter a number: "))
2 if number % 2 == 0:
3     print(f"{number} is an even number")
4 else:
5     print(f"{number} is an odd number")
```

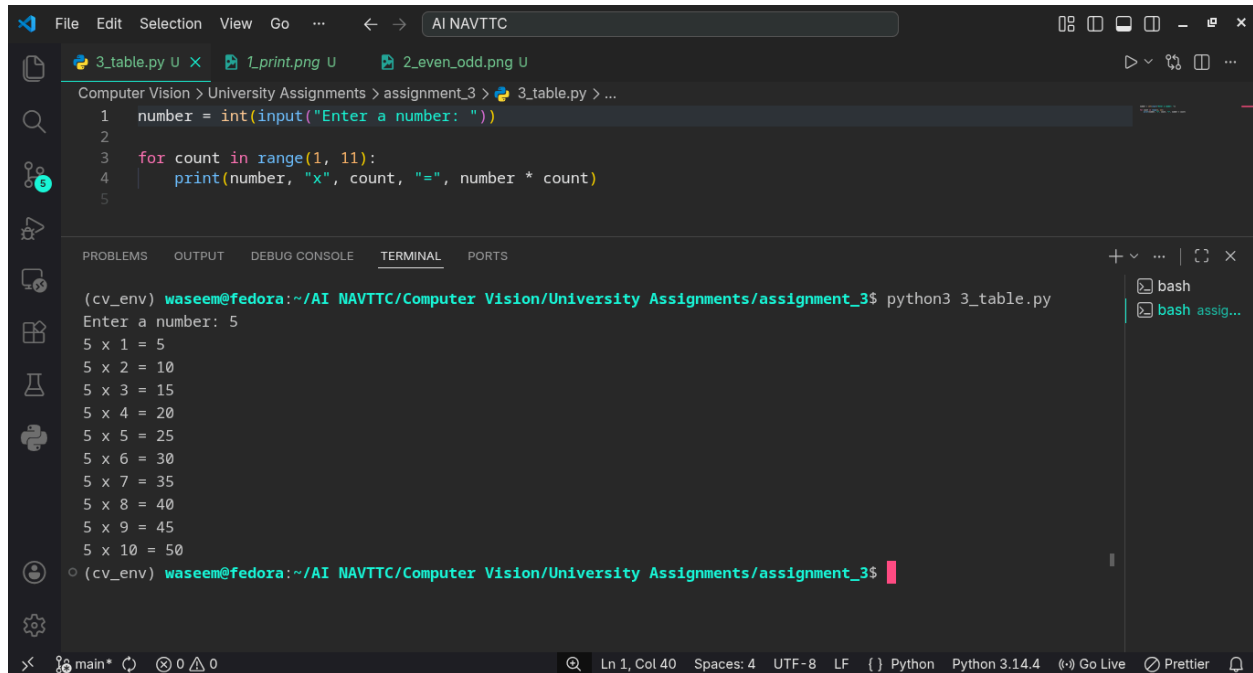
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
```

```
bash - assignment_3
```

```
(cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$ python3 2_even_odd.py
Enter a number: 8
8 is an even number
(cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$ python3 2_even_odd.py
Enter a number: 7
7 is an odd number
(cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$
```

Ln 5, Col 40 Spaces: 4 UTF-8 LF {} Python Python 3.14.4 Go Live Prettier

Task 3: Print the multiplication table of a given number:



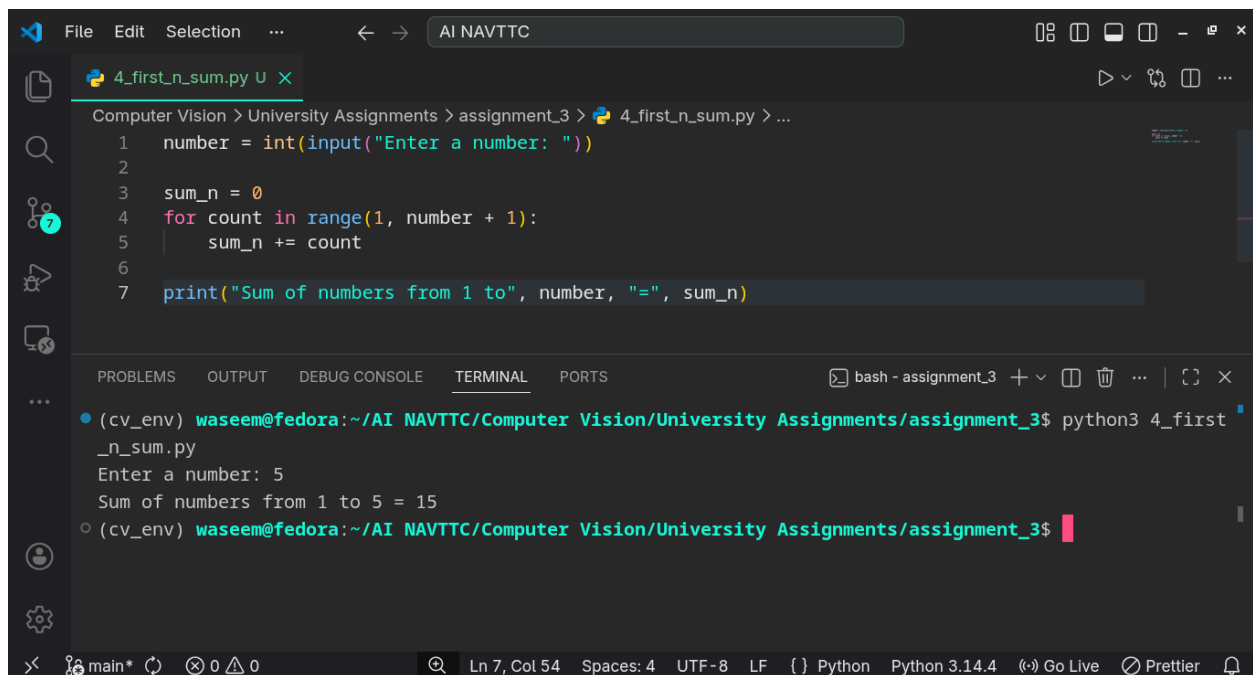
The screenshot shows a VS Code editor with a file named `3_table.py` open. The code is as follows:

```
1 number = int(input("Enter a number: "))
2
3 for count in range(1, 11):
4     print(number, "x", count, "=", number * count)
5
```

The terminal window shows the execution of the script:

```
(cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$ python3 3_table.py
Enter a number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
(cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$
```

Task 4: Print the sum of first n natural numbers:



The screenshot shows a VS Code editor with a file named `4_first_n_sum.py` open. The code is as follows:

```
1 number = int(input("Enter a number: "))
2
3 sum_n = 0
4 for count in range(1, number + 1):
5     sum_n += count
6
7 print("Sum of numbers from 1 to", number, "=", sum_n)
```

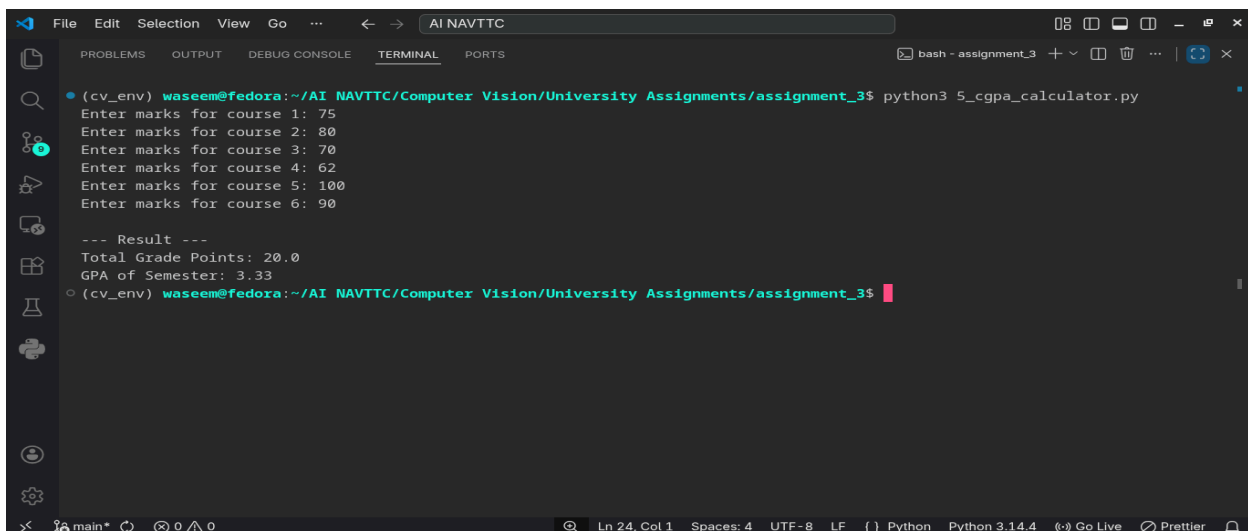
The terminal window shows the execution of the script:

```
(cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$ python3 4_first_n_sum.py
Enter a number: 5
Sum of numbers from 1 to 5 = 15
(cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$
```

Task 5: GP and GPA calculator:

Computer Vision > University Assignments > assignment_3 > 5_cgpa_calculator.py > ...

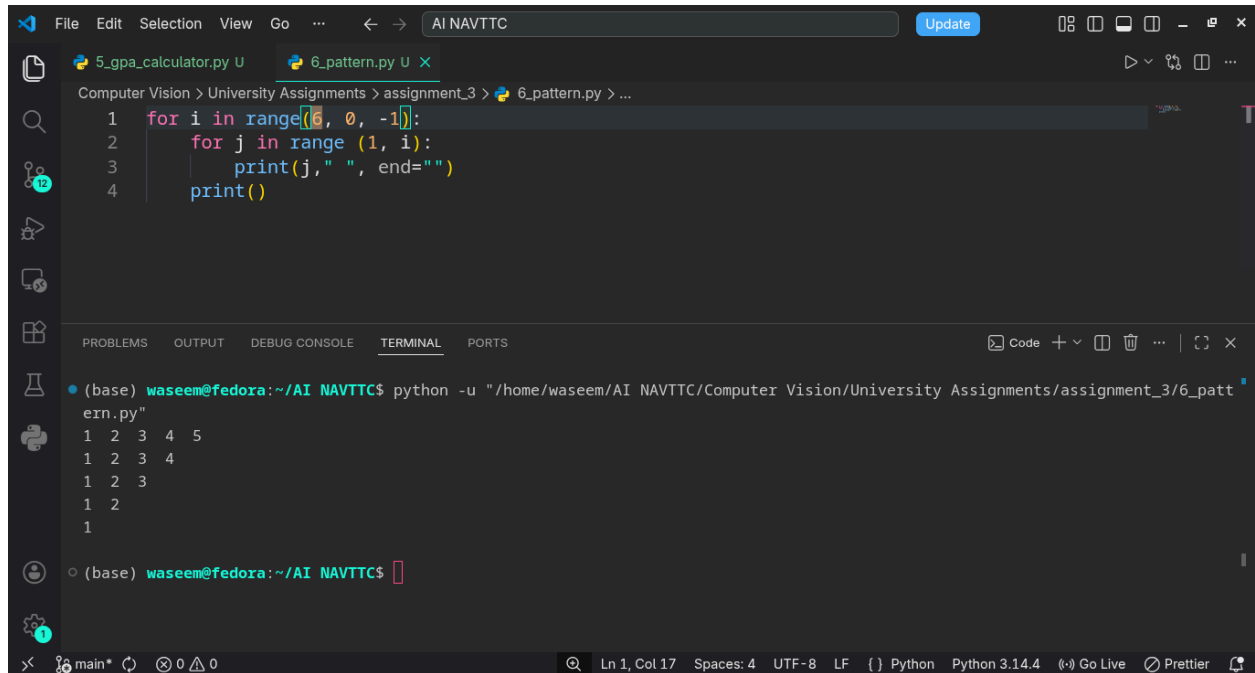
```
1 marks = []
2
3 for i in range(6):
4     mark = int(input(f"Enter marks for course {i+1}: "))
5     marks.append(mark)
6
7 total_gp = 0
8
9 for m in marks:
10     if m >= 80:
11         gp = 4.0
12     elif m >= 70:
13         gp = 3.0
14     elif m >= 60:
15         gp = 2.0
16     elif m >= 50:
17         gp = 1.0
18     else:
19         gp = 0.0
20
21     total_gp += gp
22
23 gpa = total_gp / 6
24
25 print("\n--- Result ---")
26 print("Total Grade Points:", total_gp)
27 print("GPA of Semester:", round(gpa, 2))
```



```
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
bash - assignment_3
(cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$ python3 5_cgpa_calculator.py
Enter marks for course 1: 75
Enter marks for course 2: 80
Enter marks for course 3: 70
Enter marks for course 4: 62
Enter marks for course 5: 100
Enter marks for course 6: 90

--- Result ---
Total Grade Points: 20.0
GPA of Semester: 3.33
(cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$
```

Task 6: Pattern Problem:



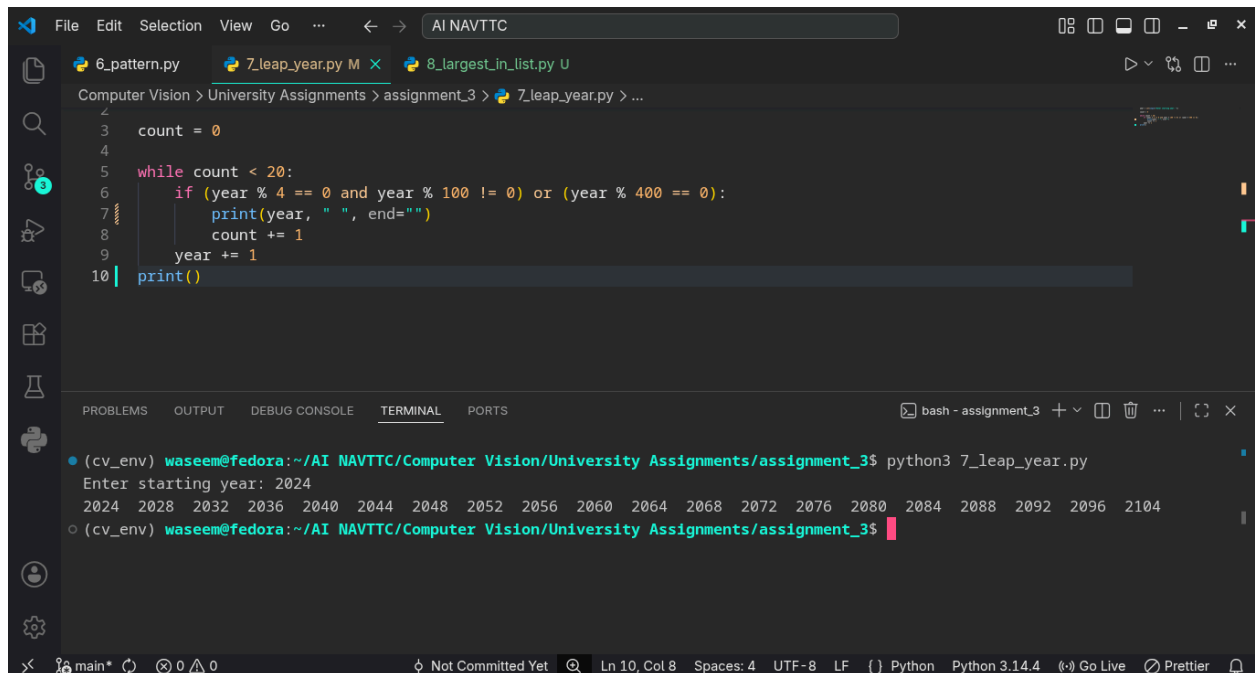
The screenshot shows the VS Code editor with a file named `6_pattern.py` open. The code is a nested loop that prints a pattern of numbers. The terminal shows the output of running the script.

```
1 for i in range(6, 0, -1):
2     for j in range(1, i):
3         print(j, " ", end="")
4     print()
```

Terminal output:

```
(base) waseem@fedora:~/AI NAVTTC$ python -u "/home/waseem/AI NAVTTC/Computer Vision/University Assignments/assignment_3/6_pattern.py"
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

Task 7: Leap Year:



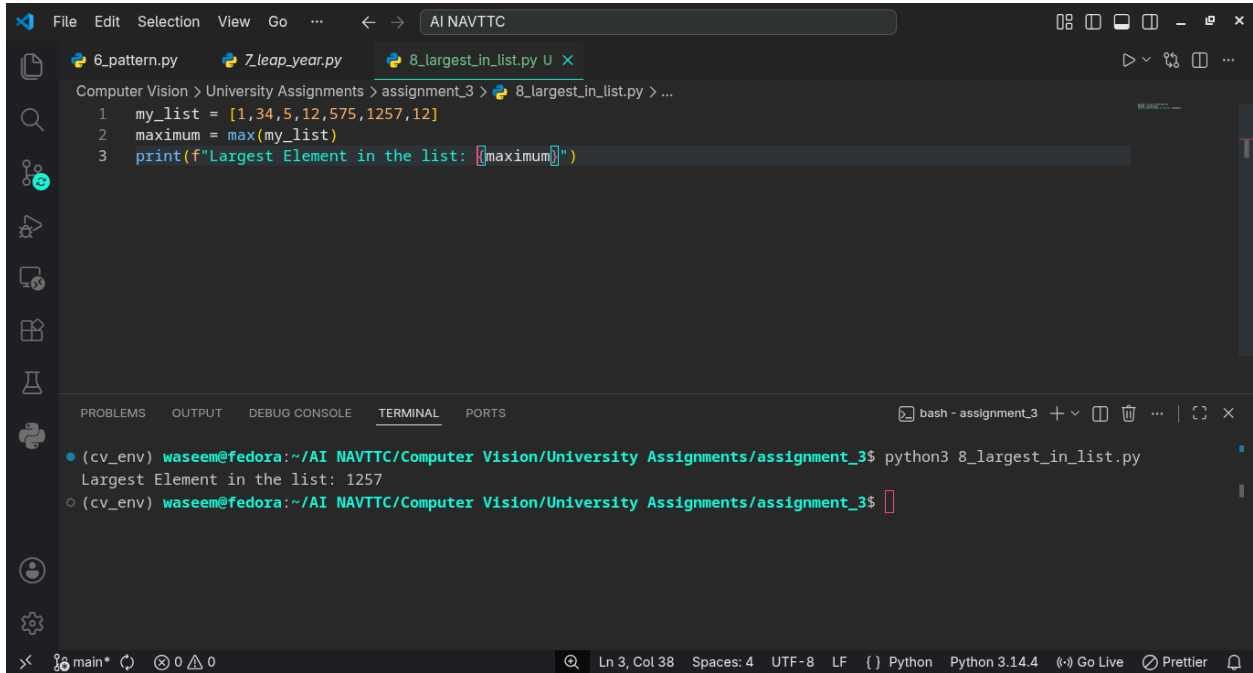
The screenshot shows the VS Code editor with a file named `7_leap_year.py` open. The code is a while loop that prints leap years from 2024 to 2104. The terminal shows the output of running the script.

```
3 count = 0
4
5 while count < 20:
6     if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
7         print(year, " ", end="")
8         count += 1
9     year += 1
10 print()
```

Terminal output:

```
(cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$ python3 7_leap_year.py
Enter starting year: 2024
2024 2028 2032 2036 2040 2044 2048 2052 2056 2060 2064 2068 2072 2076 2080 2084 2088 2092 2096 2104
```

Task 8: Largest Number in List:

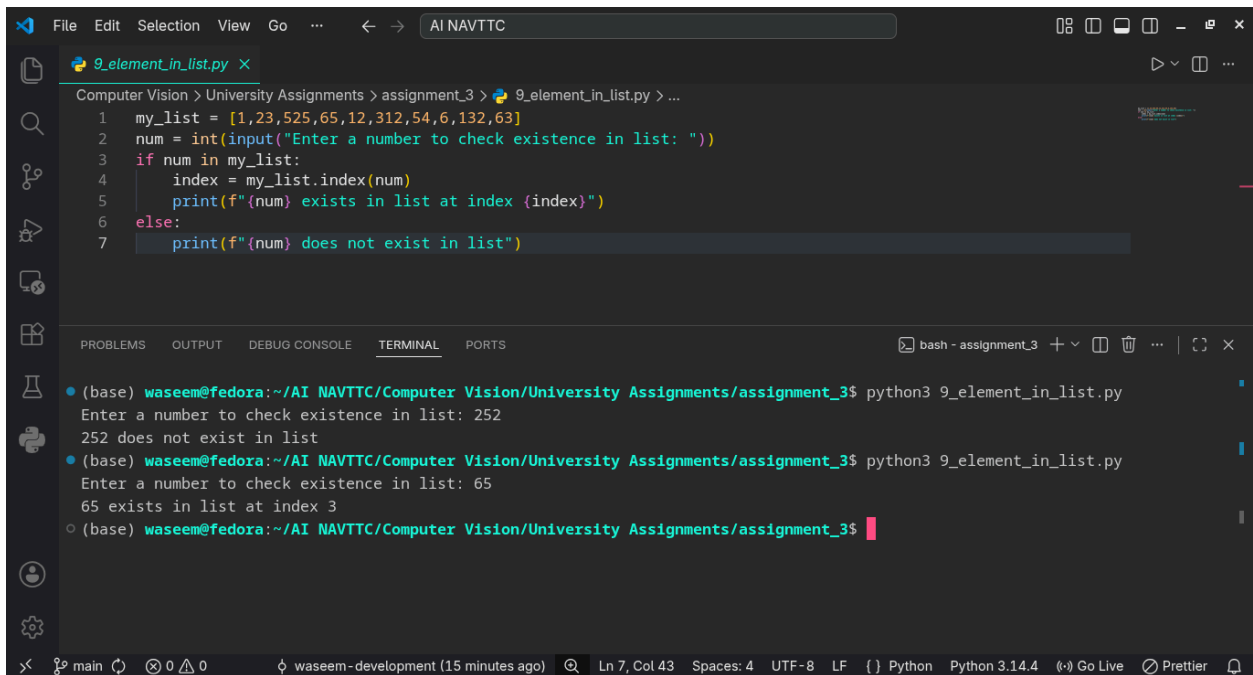


The screenshot shows a VS Code editor window with the file `8_largest_in_list.py` open. The code defines a list `my_list` and uses the `max()` function to find the largest element. The terminal output shows the script being executed and the result `1257`.

```
File Edit Selection View Go ... AI NAVTTC
6_pattern.py Z leap_year.py 8_largest_in_list.py U X
Computer Vision > University Assignments > assignment_3 > 8_largest_in_list.py > ...
1 my_list = [1,34,5,12,575,1257,12]
2 maximum = max(my_list)
3 print(f"Largest Element in the list: {maximum}")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
bash - assignment_3
• (cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$ python3 8_largest_in_list.py
Largest Element in the list: 1257
○ (cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$
```

Task 9: Check if a number exists in a list:

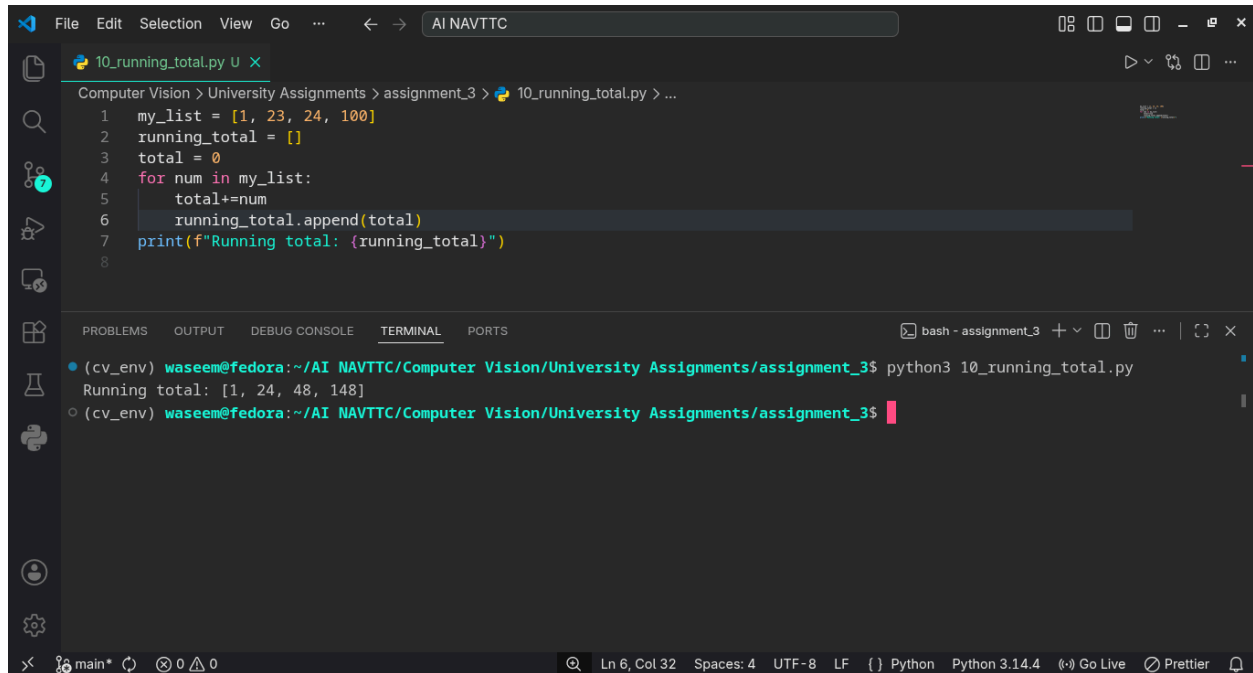


The screenshot shows a VS Code editor window with the file `9_element_in_list.py` open. The code prompts the user to enter a number and checks if it exists in the list `my_list`. The terminal output shows two test cases: `252` (does not exist) and `65` (exists at index 3).

```
File Edit Selection View Go ... AI NAVTTC
9_element_in_list.py X
Computer Vision > University Assignments > assignment_3 > 9_element_in_list.py > ...
1 my_list = [1,23,525,65,12,312,54,6,132,63]
2 num = int(input("Enter a number to check existence in list: "))
3 if num in my_list:
4     index = my_list.index(num)
5     print(f"{num} exists in list at index {index}")
6 else:
7     print(f"{num} does not exist in list")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
bash - assignment_3
• (base) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$ python3 9_element_in_list.py
Enter a number to check existence in list: 252
252 does not exist in list
• (base) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$ python3 9_element_in_list.py
Enter a number to check existence in list: 65
65 exists in list at index 3
○ (base) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$
```

Task 10: Running Total of a List:



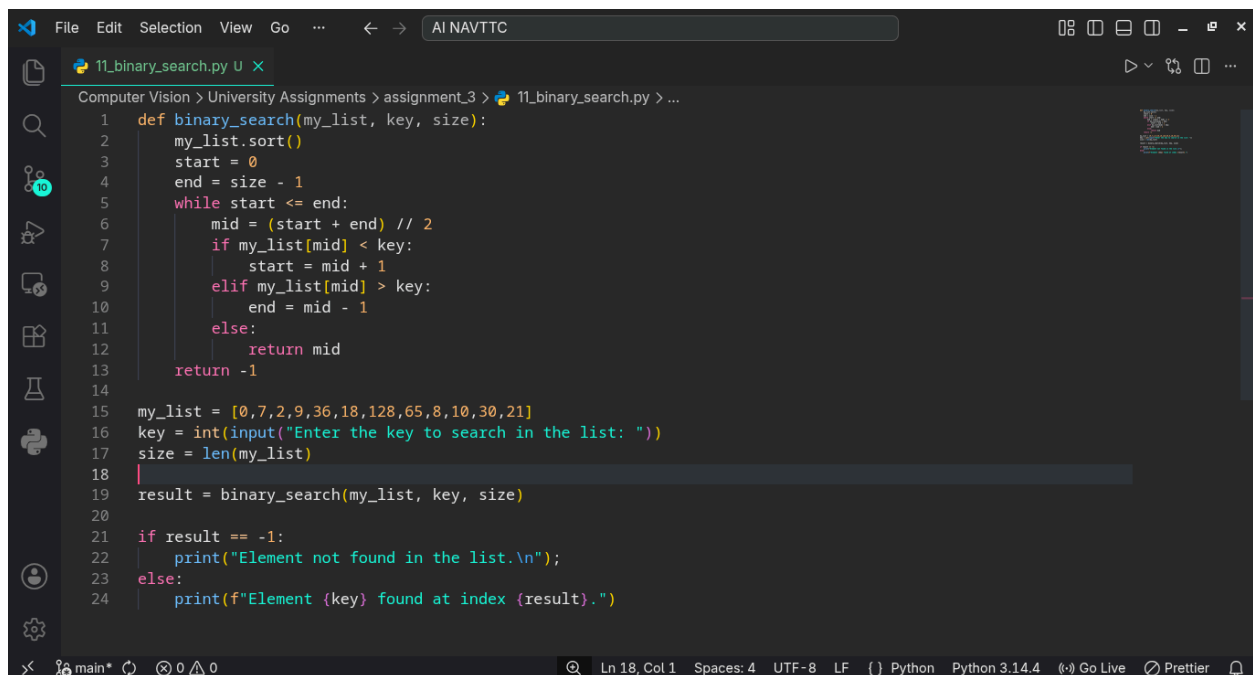
The screenshot shows a code editor with a file named `10_running_total.py`. The code implements a function to calculate the running total of a list. The terminal output shows the execution of the script, which prints the running total at each step.

```
1 my_list = [1, 23, 24, 100]
2 running_total = []
3 total = 0
4 for num in my_list:
5     total += num
6     running_total.append(total)
7 print(f"Running total: {running_total}")
8
```

Terminal Output:

```
bash - assignment_3
(waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3) python3 10_running_total.py
Running total: [1, 24, 48, 148]
```

Task 11: Binary Search Algorithm:



The screenshot shows a code editor with a file named `11_binary_search.py`. The code implements a binary search algorithm. The terminal output shows the execution of the script, which prompts the user to enter a key to search for and prints the result.

```
1 def binary_search(my_list, key, size):
2     my_list.sort()
3     start = 0
4     end = size - 1
5     while start <= end:
6         mid = (start + end) // 2
7         if my_list[mid] < key:
8             start = mid + 1
9         elif my_list[mid] > key:
10            end = mid - 1
11        else:
12            return mid
13    return -1
14
15 my_list = [0,7,2,9,36,18,128,65,8,10,30,21]
16 key = int(input("Enter the key to search in the list: "))
17 size = len(my_list)
18
19 result = binary_search(my_list, key, size)
20
21 if result == -1:
22     print("Element not found in the list.\n");
23 else:
24     print(f"Element {key} found at index {result}.")
```

Terminal Output:

```
bash - assignment_3
(waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3) python3 11_binary_search.py
Enter the key to search in the list: 10
Element 10 found at index 10.
```

The image shows a Visual Studio Code (VS Code) window with a terminal open. The terminal title is "AI NAVTTC". The terminal output shows the execution of a Python script named "11_binary_search.py". The script prompts the user to enter a key to search in the list. In the first run, the user enters "65", and the output is "Element 65 found at index 10.". In the second run, the user enters "888", and the output is "Element not found in the list.". The terminal prompt is "(cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3\$". The VS Code interface includes a menu bar (File, Edit, Selection, View, Go, ...), a toolbar, and a sidebar with icons for Explorer, Search, Run and Debug, and Extensions. The status bar at the bottom shows "main*", "Ln 18, Col 1", "Spaces: 4", "UTF-8", "LF", "Python", "Python 3.14.4", "Go Live", "Prettier", and a bell icon.

```
File Edit Selection View Go ... ← → AI NAVTTC
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
• (cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$ python3 11_binary_search
.py
Enter the key to search in the list: 65
Element 65 found at index 10.
• (cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$ python3 11_binary_search
.py
Enter the key to search in the list: 888
Element not found in the list.
○ (cv_env) waseem@fedora:~/AI NAVTTC/Computer Vision/University Assignments/assignment_3$
main* 0 0 0 Ln 18, Col 1 Spaces: 4 UTF-8 LF {} Python Python 3.14.4 Go Live Prettier
```